

CARDIFF WEATHER CENTRE, United Kingdom

WMO: 037170

Lat: 51.483N

Lon: 3.183W

Elev: 171

StdP: 14.61

Time Zone: 0.00 (EUW)

Period: 94-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.6	32.5	21.1	15.9	36.3	23.7	18.0	37.6	31.8	49.1	29.1	49.2	9.0	60	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.5	79.7	65.4	76.1	63.7	73.1	62.2	66.9	76.4	65.1	73.4	63.6	70.7	7.7	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.4	87.8	69.9	62.0	83.6	68.2	60.7	79.7	66.9	31.4	76.7	30.1	73.9	29.0	70.7	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.1	22.8	20.1	DB	26.4	84.4	1.7	2.6	25.2	86.2	24.3	87.8	23.3	89.2	22.1	91.1	(4)
(5)			WB	24.5	70.3	1.9	2.5	23.2	72.1	22.1	73.6	21.0	75.0	19.6	76.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.0	43.3	44.0	46.5	49.9	55.3	60.3	64.0	64.9	60.0	54.4	48.6	44.0	(6)	
	DBStd	8.94	5.07	4.94	4.69	4.59	4.85	4.28	4.15	4.55	3.66	4.43	4.69	5.49	(7)	
	HDD50	860	212	173	124	56	8	0	0	0	0	13	80	194	(8)	
	HDD65	4500	674	589	573	454	303	155	68	58	157	328	491	650	(9)	
	CDD50	1947	2	4	15	51	172	308	435	461	300	151	39	9	(10)	
	CDD65	114	0	0	0	0	2	13	39	54	6	0	0	0	(11)	
	CDH74	546	0	0	0	2	13	75	191	245	20	0	0	0	(12)	
	CDH80	93	0	0	0	0	0	12	26	54	1	0	0	0	(13)	
(14) Wind	WSAvg	9.4	11.0	10.5	10.1	9.5	9.6	8.7	8.0	7.5	8.5	10.0	8.8	10.2	(14)	
(15) Precipitation	PrecAvg	42.2	4.4	3.3	3.3	2.6	2.6	2.7	2.7	3.5	3.9	4.3	4.6	4.7	(15)	
	PrecMax	55.3	8.5	7.2	7.9	5.6	6.6	6.4	6.5	9.9	8.6	9.5	9.8	11.0	(16)	
	PrecMin	31.0	0.6	0.1	0.4	0.3	0.2	0.4	0.6	0.4	0.8	0.5	1.5	1.4	(17)	
	PrecStd	5.4	2.2	2.0	1.8	1.3	1.5	1.7	1.6	2.1	2.0	2.2	2.2	2.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	56.9	61.3	69.5	75.9	81.1	83.1	85.7	76.9	66.7	59.7	56.4	(19)	
		MCWB	51.2	52.2	50.8	55.2	60.7	64.4	65.5	67.3	64.4	59.3	56.3	53.4	(20)	
	2%	DB	53.2	54.0	57.5	63.3	71.4	75.4	79.2	80.3	72.3	63.9	58.2	55.0	(21)	
		MCWB	50.1	50.4	49.9	52.9	59.6	62.1	64.8	67.0	62.1	57.7	55.1	52.5	(22)	
	5%	DB	51.5	52.2	54.8	59.2	67.0	71.6	75.4	75.8	68.8	62.0	56.7	53.5	(23)	
		MCWB	48.7	48.5	48.6	50.6	57.6	60.2	63.5	64.6	59.8	57.2	53.7	50.8	(24)	
	10%	DB	50.3	50.7	52.9	56.6	63.5	68.0	72.0	72.6	66.2	60.7	55.4	51.8	(25)	
		MCWB	47.6	47.4	48.3	49.4	55.2	58.3	62.2	62.8	58.5	56.5	52.3	49.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.1	53.2	53.3	56.7	63.5	66.2	68.2	69.7	66.4	61.7	57.6	54.4	(27)	
		MCDB	54.0	55.7	56.9	67.9	72.1	77.2	78.8	81.3	75.0	64.6	58.7	55.5	(28)	
	2%	WB	50.7	51.2	51.7	53.9	60.5	63.3	66.1	67.7	63.4	59.9	56.1	53.3	(29)	
		MCDB	52.5	53.4	54.5	60.5	68.6	72.8	76.6	77.3	69.5	62.1	57.5	54.8	(30)	
	5%	WB	49.4	49.6	50.4	52.1	58.3	61.3	64.5	65.8	61.8	58.5	54.5	51.3	(31)	
		MCDB	51.2	51.6	53.1	57.4	65.8	69.1	72.8	73.6	66.3	61.1	56.3	52.9	(32)	
	10%	WB	48.1	48.0	49.3	50.5	56.2	59.7	63.0	64.1	60.4	57.2	52.9	49.5	(33)	
		MCDB	50.1	50.3	52.1	55.0	62.5	66.3	70.4	70.6	64.5	60.1	55.0	51.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.7	9.8	11.8	12.6	13.0	13.1	13.5	12.0	9.8	8.5	7.6	(35)	
		MCDBR	8.5	9.2	13.1	15.7	18.4	18.9	19.5	18.3	16.0	10.2	8.5	7.6	(36)	
	5% WB	MCWBR	7.6	7.4	8.0	8.4	9.7	9.1	9.3	8.4	8.2	6.9	7.1	7.5	(37)	
		MCWBR	8.1	8.4	10.0	13.4	16.7	16.2	16.2	16.6	13.4	8.8	7.8	7.5	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.331	0.345	0.377	0.397	0.402	0.393	0.391	0.377	0.369	0.355	0.330	0.321	(40)	
		taud	2.346	2.372	2.287	2.254	2.281	2.341	2.350	2.406	2.422	2.425	2.400	2.323	(41)	
	Ebn at Noon	Ebn at Noon	213	245	257	266	270	274	272	270	259	238	213	194	(42)	
		Edh at Noon	20	26	34	39	40	38	37	33	29	24	19	18	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	248	468	829	1276	1567	1702	1592	1319	993	562	298	193	(44)	
	RadStd	RadStd	22	57	106	146	79	180	150	116	73	49	29	24	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page